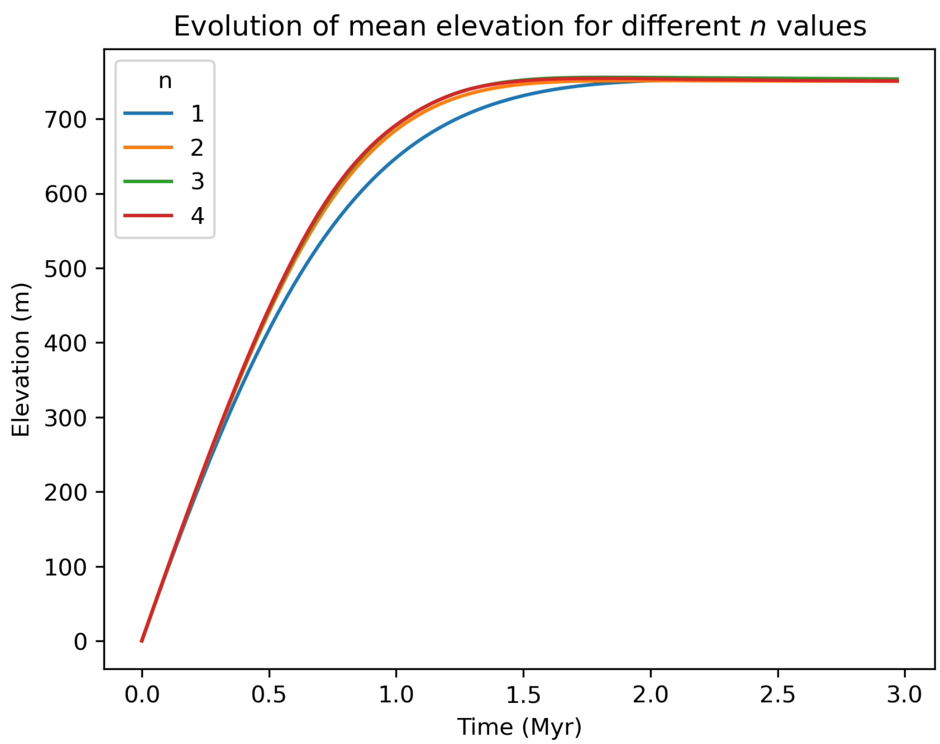
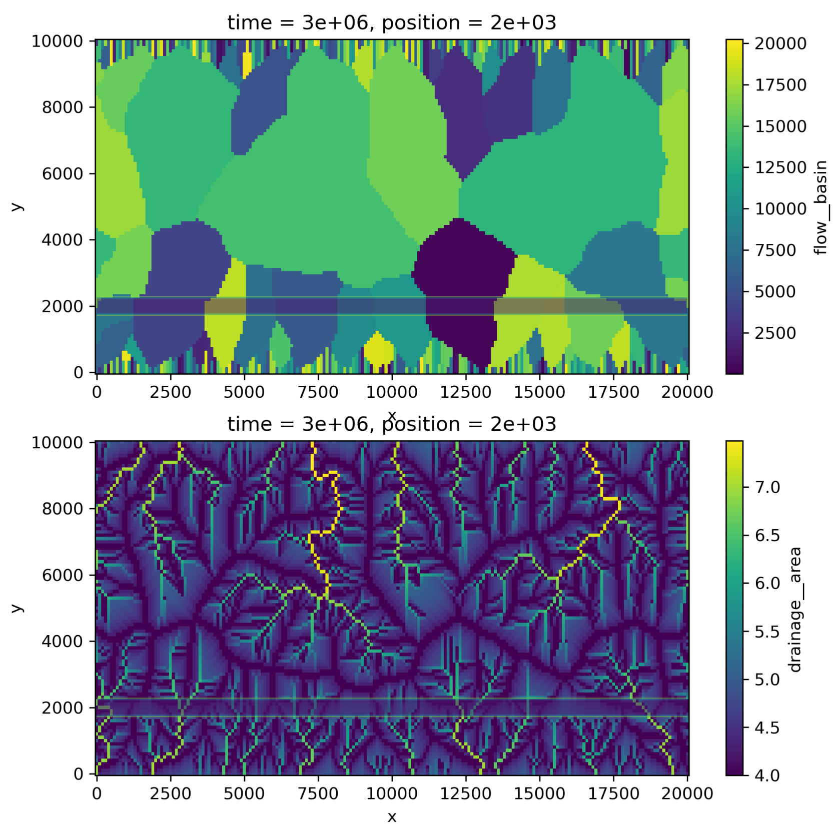
Project 1: Effect of varying the slope exponent on the growth and decay of a mountain belt

Question: Compare the duration of the growth and decay of a mountain belt. How important is the value of the slope exponent, n?
Suggested Method: Perform a simple experiment to create a mountain belt until it reaches steady-state. Estimate the (e-folding) time to steady-state. Perform a series of experiments varying the value of the slope exponent, n, in the stream power law ($n = 1, 2, 3, 4$). Adjust the value of the erodibility (K_f) such that the steady-state mean mountain height is the same in all experiments. In a second set of experiments, double the duration of the experiments and set the uplift rate to zeros in the second-half of the experiment, simulating the end of a collision between two continents. For each experiment, estimate the (e-folding) time for the decay phase of the mountain belt.



Project 2: Effect of a vertical dyke on drainage re-organization

Question: What is the effect of a vertical dyke (strong vertical layer) on the geometry of the drainage network?
Suggested Method: Perform series of experiments varying the dyke position and thickness and analyse the drainage network and the catchment distribution and geometry. Make sure that only the two boundaries parallel to the dyke are set at base level ("fixed_value") and the other two as cyclic boundaries ("looped")



Project 3: Introducing a threshold area for fluvial erosion

Question: what are the effects on a landscape to restrict stream power law erosion to areas above a minimum drainage area, a_{min}?
Suggested Method: Develop a new process that contribute to the group variable "erosion" and takes as an input the "erosion" variable out of the spl process and the drainage area. In this process, the contribution to erosion will be 0 where drainage area is larger than a_{min} and -erosion (from spl) otherwise. Perform a series of experiments varying the value of a_{min}.

```
from fastscape.processes import DrainageArea, StreamPowerChannel

@xs.process
class ThresholdArea:
    area = xs.variable(description="Minimum drainage area
                                    for channel erosion")
    drainage_area = xs.foreign(DrainageArea, 'area')
    erosion_spl = xs.foreign(StreamPowerChannel, 'erosion',
                             intent='in')
    erosion = xs.variable(
        dims=("y", "x"),
        intent="out",
        groups="erosion",
    )

    def run_step(self):
        self.erosion = ... YOUR CODE ...
```

Project 4: Disentangling climate from tectonic forcing on the growth of a mountain belt

Question: During the evolution of a mountain belt, can you differentiate between an increase in uplift rate or precipitation rate?

Suggested Method: Perform an experiment made of two episodes. In the first episode, take a mountain to topographic steady-state and in the second episode, bring it to another steady-state by doubling the uplift rate. Perform a second experiment by decreasing the precipitation rate in the second episode. Choose the amplitude of this change to obtain the same final steady-state topography than in the first experiment.

Analyze the topography that you obtain in each case and compare them to each other. Compare also the time evolution of the topography. You should also perform this comparison with a slope exponent $n = 2$ (pay attention to also change the value of m so that the ratio m/n remains constant and adjust the value of K_f so that you obtain a similar mean topography at steady-state with $n = 1$ or $n = 2$). .